SOA Rocks B2B Foundation

Enterprise Service Networks Evolve

About the Speaker

Speaker Name:
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Company:
MomentumSI

Title:
VP, Business Innovations & Transformations

Background:
Ed Vazquez leads the Enterprise Transformation Practice. In his more than a decade of Web Application Management (including six years of SOA and Web Service Implementation Management), Ed has led major programs to transform enterprise Business and IT organizations with Sprint, Pfizer, and other companies. Prior to focusing on Enterprise Transformation and Innovation, Ed developed significant subject matter expertise in the SOA Lifecycle, SOA Governance, and Business Driven SOA. Ed graduated with both his Masters Degree and Bachelors Degree from the University of Kansas.

Agenda

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- About MomentumSI
- About SOA (and B2B)
- The Opportunity: The Virtualized Enterprise
- The External Service Network
- Industry Examples

MomentumSI

Consulting Services

About MomentumSI

MomentumSI is a boutique, 70-person, **consulting** company focusing on business software solutions.

Areas of expertise:

- Enterprise Architecture & SOA
- Software Development Services
- Training & Mentoring

Founded	– 1997, privately held
HQ	– Austin, TX
Regional	– San Francisco, D.C., New York

MomentumSI Clients

COX COMMUNICATIONS
Sabre Travel Network
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Sun microsystems
AMD
intel.
3M
AmerisourceBergen Specialty Group
Armstrong
The Walt Disney Company

MomentumSI Consulting

- Outsourced SOA Foundation Programs
  - Enterprise Architecture, Training, Governance, Service Portfolio Management & Analysis
- Project Mentors for SOA
  - SOA Project Manager Professionals, Service Analysts, Service Designers, Service Developers, Integration Analysts, Tool (ESB, Registry / Repository, etc) Specialists, Composite Application Specialists
- Specialty Programs
  - External Service Networks (B2B Hubs), SOA based Master Data Management, Business Process Modeling (BPM)

About SOA

- Service Oriented Architecture and B2B

What is SOA

- SOA = Service Oriented Architecture
- Services are the new unit of work (Schneider paper)

- Core Principles of SOA
  - Services share a formal contract.
  - Services are loosely coupled.
  - Services are autonomous.
  - Services abstract underlying logic.

- Other Principles of SOA
  - Services are discoverable.
  - Services are reusable.
  - Services are composable.
  - Services are stateless.

  - *Thomas Erl, *Service-Oriented Architecture: Concepts, Technology, and Design*.

- These core principles are inherent to defining the next generation of Trading Partner behaviors.
  - Trading Partners in B2B interactions are, now, able to research, discover, and integrate with services from one or more companies, themselves, to compose new services and new composite applications by consuming reusable services and service contracts (agreeing to use services for specific functions that are made available by the provider organization).

6 Pillars of SOA

- SOA Model
  - Model defines what SOA means to the enterprise and the what work is supported in an enterprise SOA program or initiative.
- SOA Organization
  - Organization defines who supports the functions involved with SOA.
- SOA Process
  - Processes define when services or work is performed.
- SOA Portfolio
  - Portfolio defines service capabilities or why services have been created.
- SOA Technology
  - Technologies and Standards define how service-orientation will work.
- SOA Tools
  - Tools define where services are created and configured to support SOA principles.

- B2B models are being impacted by the new SOA models of doing business, the new organizations that are emerging in enterprises, and the new processes that are resulting from tools and technology capabilities that are allowing portfolios of services to be created to support the enterprise “doing business” with its partners differently.

SOA Operational Lifecycle

- SOA Programs are adopting Lifecycle Models that clearly articulate that integrations are “products” and “assets” of an enterprise.
  - These “products” need to be managed to maximize Return On Investment (platform or program) and Return On Asset (service or integration).
  - These products are being offered as incentives to new and existing trading partners.
  - New products and services are, consequently, now feasible because of SOA actualization in the B2B space.
- The SOA Operational Lifecycle Model illustrates how services and integrations are marketed to trading partners, defined for trading partners to understand, made discoverable to trading partners, planned, designed, developed or configured for trading partner consumption, supported and reported on (for analysis).

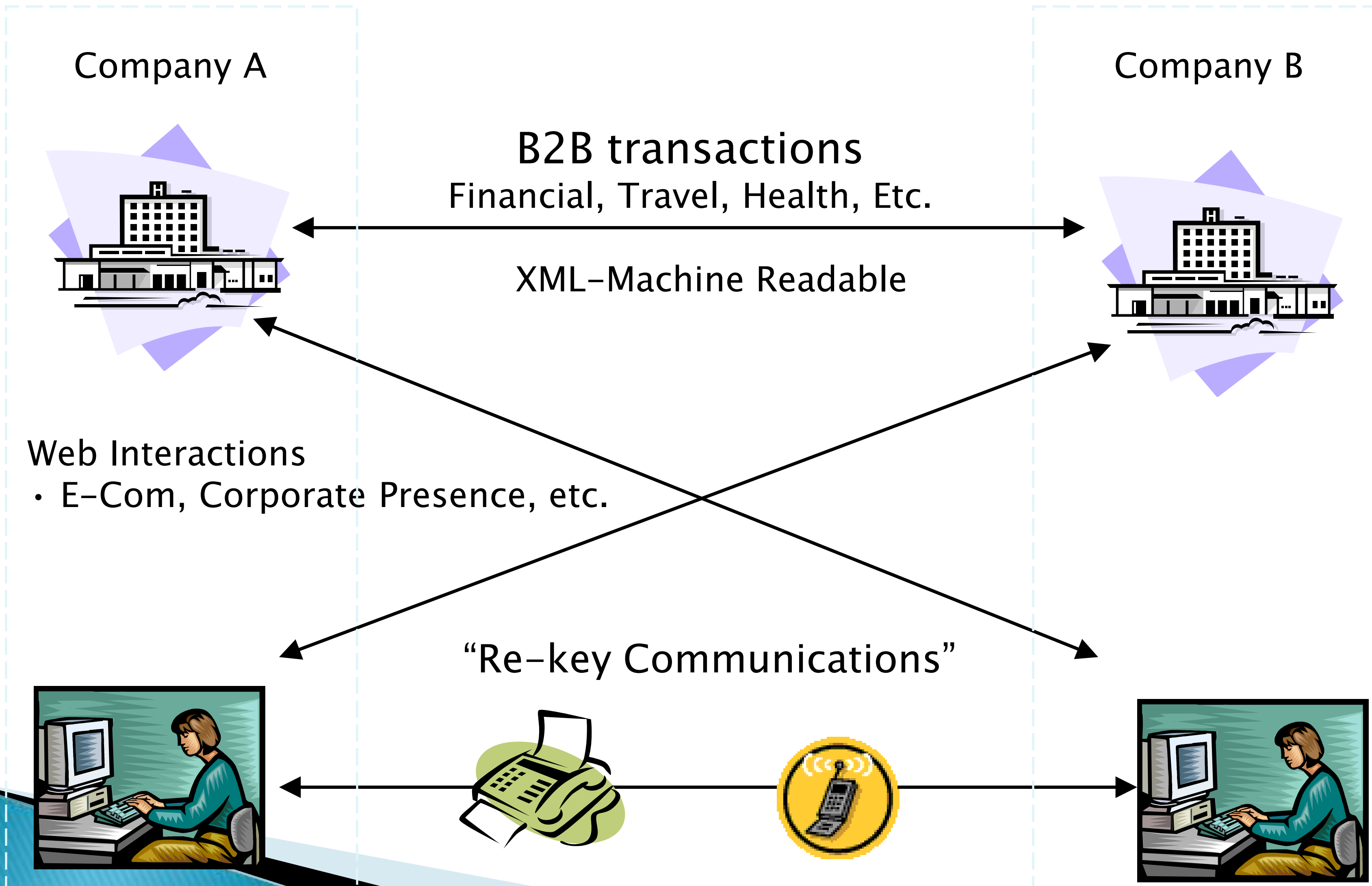
The Opportunity

The Virtualized Enterprise

SOA Today

- Many organizations have successful SOA programs underway today.
- The services that are being created can be consumed by internal applications (A2A) as well as external organizations (B2B).
- The value of a service is increased as it is used. An opportunity exists to extend the reach of the services to public and private communities of interest.

Traditional B2B Integration



B2B Integration

1970's →

Flat File Transfer

ANSI X.12 EDI

AS1 / AS2

Focus is on batches of data & common data structures

1997 →

XML & Industry Schemas

Basic Web Services

Advanced Web Services & Basic Process Integration

Focus is on ad-hoc data structures, real time exchanges & decreased integration time

2010 →

Full Process Integration

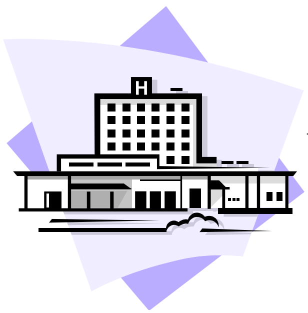
Peer-to-Peer Collaboration

Ad hoc Process & P2P Integrations

Focus in on complete process integration in real time.

Integration Vehicles

Company A



Phone / Fax

FTP Flat Files

Partner Web Portal

EDI

SOA &
Web Services

Company B

Don't force a single strategy!

Know when to use each technology…

- Real time access to information
- Reuse of existing services
- Transactions
- Subscriptions / syndications

The SOA Opportunities

- Doing SOA for the Right Reasons…

SOA Opportunity #1

- Manageable
  - The Service is bigger than an object and smaller than an application
  - The Service is the new unit of work (unit of analysis, unit of design, testing, deployment, etc.)
  - Don’t fall victim to run-away monster applications. Right size the unit.
- Impacts to B2B: Legacy functions, not just data, can be extracted and decoupled and offered to an enterprise’s trading partners resulting in agile application development for composite systems.

SOA Opportunity #2

- Synergy & Investment
  - The Service is the unit of ‘sharing’ and ‘funding’
  - SOA enables sharing software logic and data between groups.
  - SOA enables shared business processes between once disparate enterprises and organizations.
  - This enables consistency while avoid bubble-gum patches.
- Impacts to B2B: Bundles of products can be created more easily between groups and products can be resold and supported more cost-effectively. Business processes can become shared across enterprises and companies.

SOA Opportunity #3

- User-Centric Systems
  - No more 'silo applications'
  - Composite applications and mash-ups are built with the user in mind.
  - The composite is a window into the services that a person needs to do their job.
- Impact to B2B: Trading partners can create secure portals with each other through real-time integrations – to access customer data and support business functions required to support new products and services.

SOA Opportunity #4

- Process Driven
  - Process management is fundamental to business improvement
  - Silo-applications capture information about a piece of the process
  - A fundamental theme of SOA is to kill silo-applications and to separate ‘process logic’ from the clients and services
  - Impacts to B2B: Companies can create service contracts with each other and offer its own bundled, composite service and product offerings – with ONE business process, shared, resulting in lower product support costs and increased customer satisfaction. Orders are created more quickly. Orders are fulfilled more quickly. Information is validated more quickly…

SOA Opportunity #5

- Interoperability & Integration
  - Many systems were designed to accommodate the needs of the ‘funders’ but no one else
  - Integrations were a secondary thought – and often expensive and slow
  - Web Services, XML Schemas, canonical formats, mediation engines and other devices enable systems to more easily integrate.
- Impacts to B2B: With standards maturing and evolving, integrations become more practical and cost-effective than paying EDI / B2B “middlemen” and “brokers” to exchange and configure data exchanges. Organizations can afford and attain the knowledge to support their own integrations and provision integrations quickly and cost effectively.

SOA Opportunity #6

- Contracts & Service Levels
  - Traditional applications often failed to consider 'design to contract' and 'operate to SLA' as first order concerns
  - The WSDL is the outsourced contract
    - Loosely couple my vendors
  - The contract is in the contract
  - Trust is based on past performance (SLA's)
- Impacts to B2B: Integrations are reusable and not configured for each customer integration. This will significantly decrease costs to support products, going forward, while allowing more agile development in response to product changes and demands.

SOA Opportunity #7

- The Extended Enterprise
  - A value chain is a series of activities that produce some valuable output and often extend across several businesses and government agencies.
  - Applications can now be designed to work directly with those of other organizations. The boundaries of software have extended beyond a single company.
  - Organizations must know their role in a value chain and produce and consumer services accordingly.
- Impact to B2B: Companies become cooperative and learn that partnering through inexpensive and reusable integrations will result in new service offerings that support exactly what their Customers want, as they want it.

The External Service Network

Extending SOA

What is an External Service Network?

- Organizations are creating a 'capability' to extend their services to partners and other communities of interest.

- Creating this capability requires a combination of:
  - Architectural blueprints
  - New Infrastructure
  - Policies and procedures
  - Trained people
  - A service mentality (shared service center)

Operational View

External Groups

Suppliers
Customers
Information Bureaus
Distributors
Regulatory Groups
Etc.

Provides Services

ESN Team

Define virtual teams
Broker business contract
Create SLA’s

Define Service Needs
Do integrations & provisioning

ESN Help Desk
Manage external registry
Manage external security
Manage external SLA’s

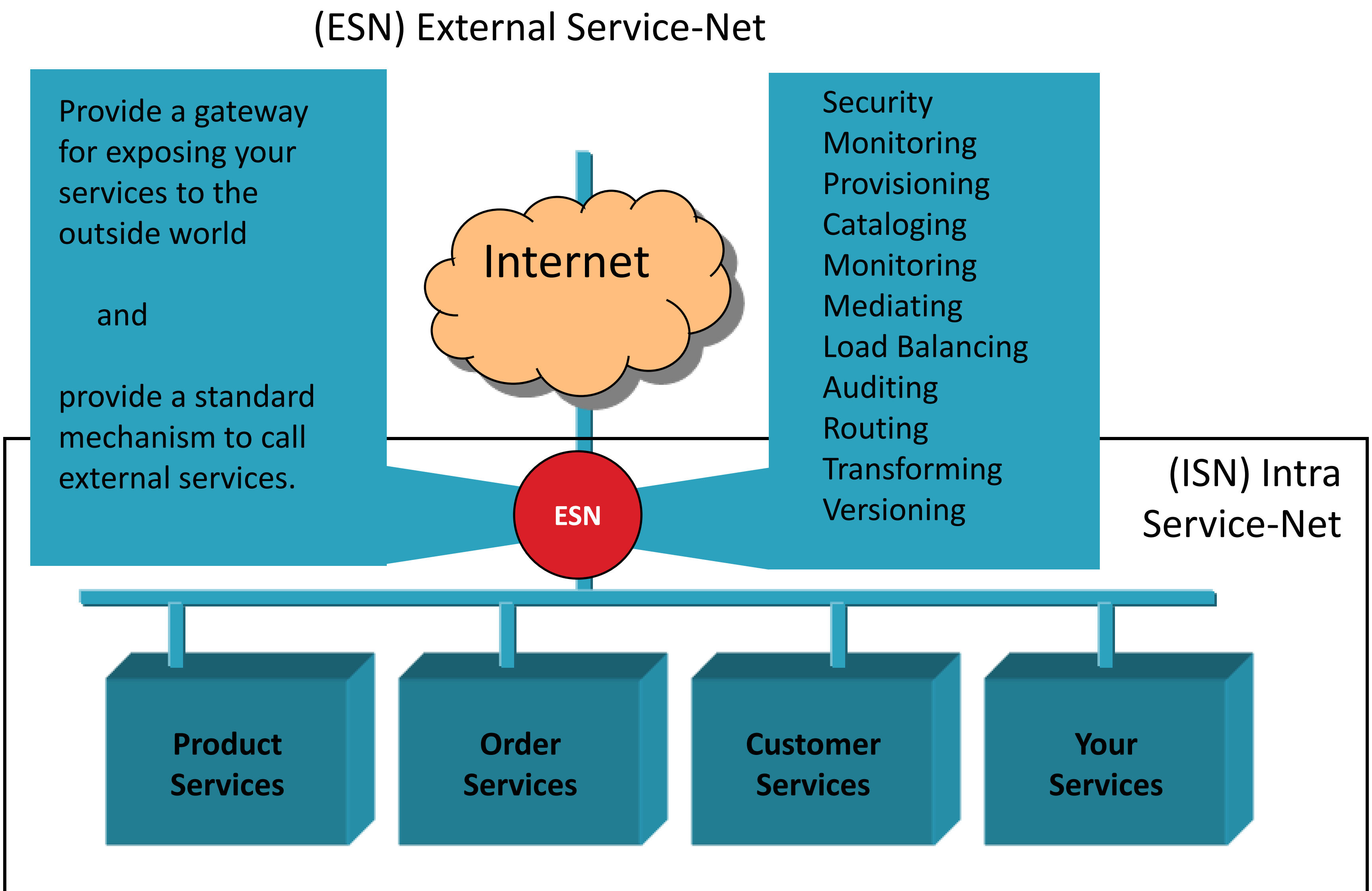
Forecast consumption demand
Facilitate changes & versions

Provides Services

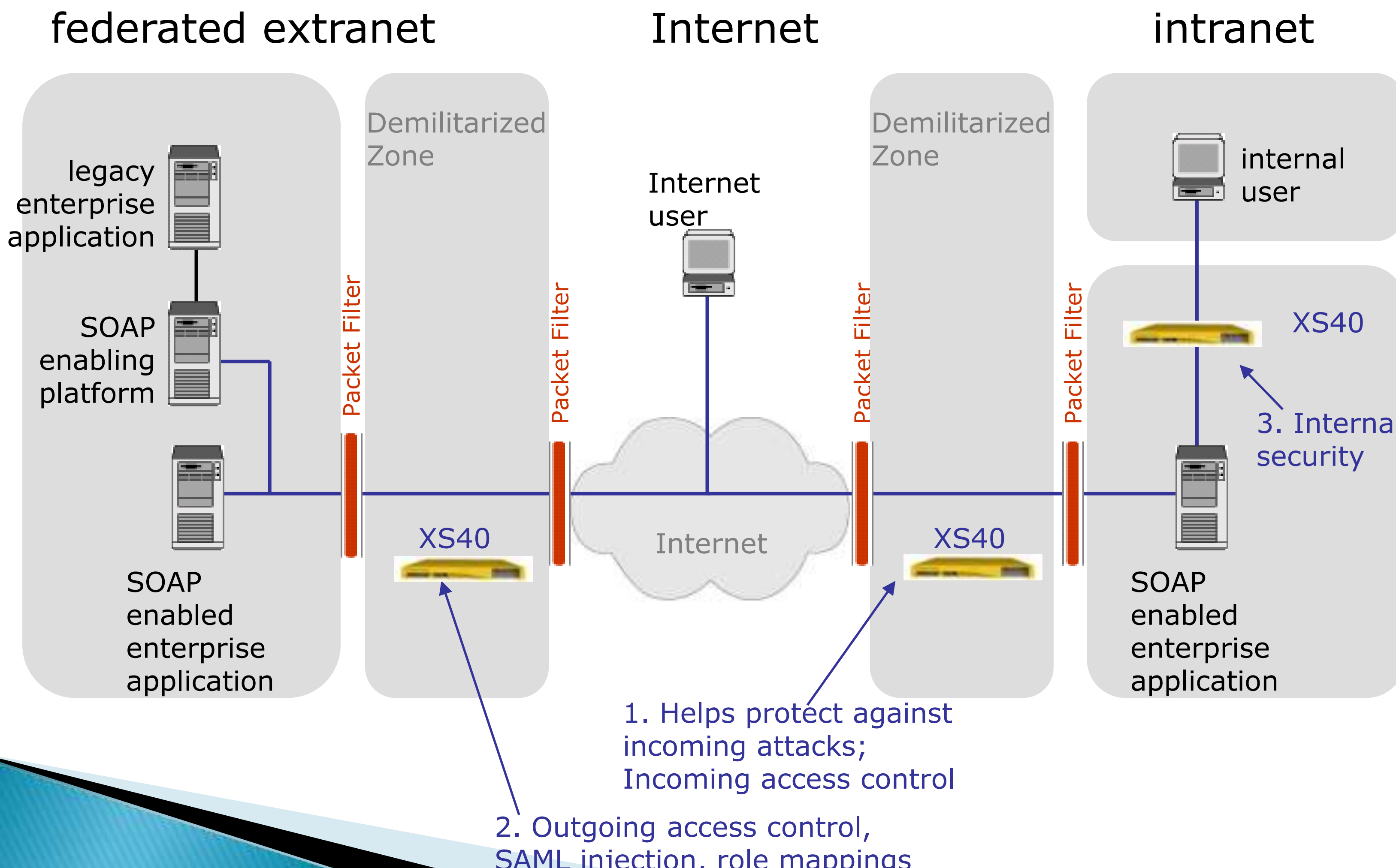
Internal Groups

Line of Business
Sales
Product
Etc.

Functional View



Physical View



Security Concerns

SOA Security is primarily concerned with:

- Issues related to network messaging
- Changing the granularity of the secured entity (message & service).

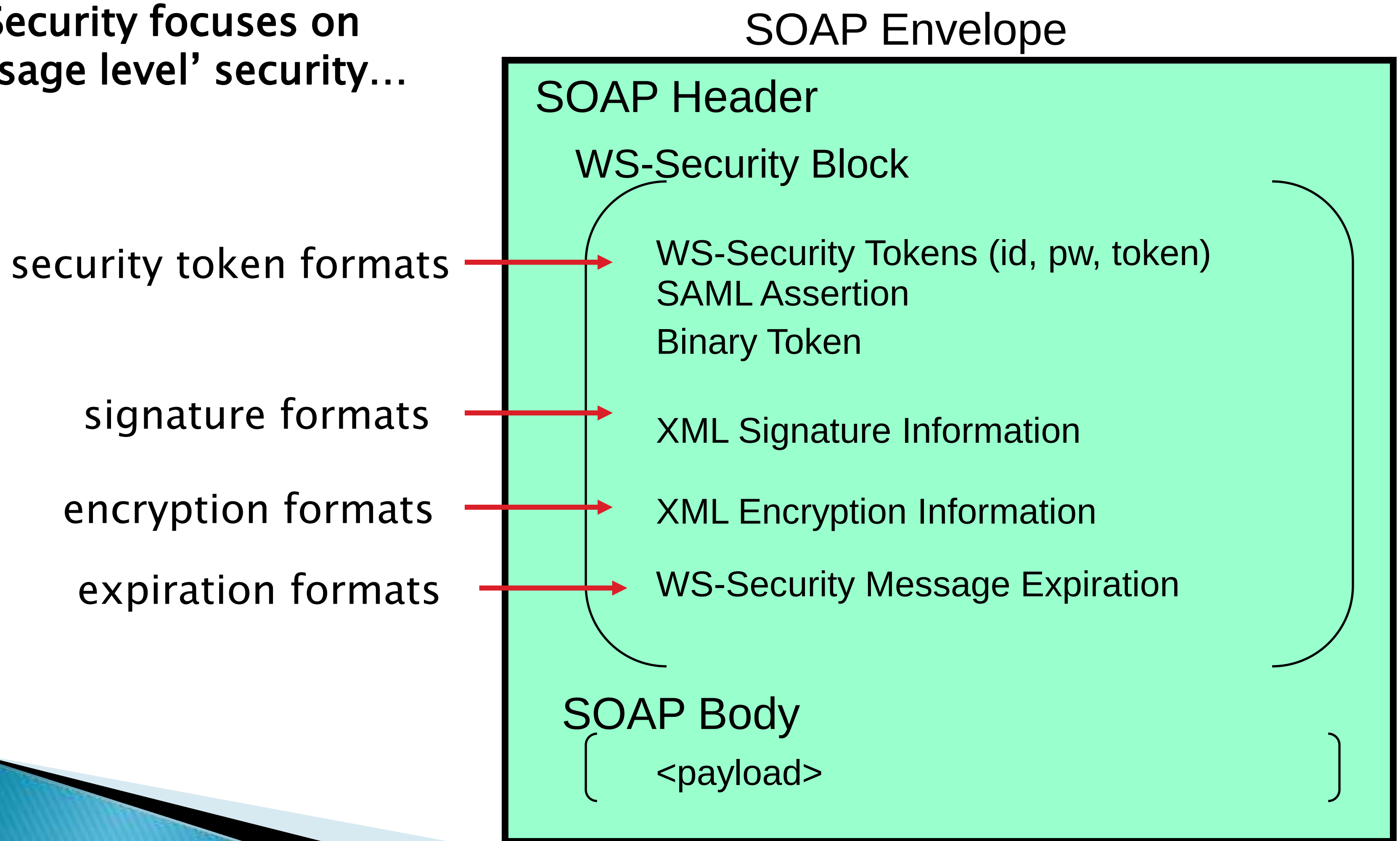
1. Identity & Access Control

2. Message Confidentiality (encrypted)
3. Message Integrity (hashed)
4. Message Authenticity (signed by originator)
5. Message Non-Repudiation (logged)

6. Cross-Enterprise Trust

Web Services Security

WS-Security focuses on ‘message level’ security…



Beyond Security

- The External Service Network must be designed to accommodate change
- Create a goal of 'Self Service SOA'
  - Remove manual non-value added activities
- Establish 'operational excellence' early
  - The ESN must be run well defined policies and procedures

Common Issues

Consuming and providing services to external parties is often more complex:

- Legal / contractual business issues
- Security over the Internet
- Security: authentication & access control
- Unknown demand from external parties
- Contractual need for SLA's
- Can't 'force hand' in versioning
- More documentation (not a cube away)

Industry Examples

» Extending Services to Partners

Travel Industry

Sample Services:

- Book a flight
- Request real time pricing
- Cancel a hotel reservation

Sabre is an information travel hub

Use a variety of mechanisms to exchange information

SOA / Web Services are one mechanism

Leverage a SOA portal for easy access

Home | My Stuff

Asset Search

Keyword

Sabre Holdings ™

Developer Resource Center

Welcome to Sabre's Developer Resource Center - the online source for all information related to *Sabre® Web Services* (SWS), the *MySabre API* (MSAPI), and *Event Notification Services* (WSE).

Sabre Web Services

Sabre Web Services is a technology offering that provides access to Sabre® Global Distribution S... of Web services (i.e., servic... ndards). *Sabre Web Services* ... duplication across ap... (More Info...)

MySabre API

MySabre API is a fea... provides travel agencies and ... capabilities needed to build applications that integrate with the *MySabre* Emulator. This exciting advancement enables you to develop applications that seamlessly integrate with the *MySabre* Emulator and improve

Uses industry standard schemas (Open Travel Alliance)

Easy access to service catalog

Provides online documentation

What's New

- OTA_AirScheduleLLSRQ (1.1.1)
- QMoveLLSRQ (1.0.1)
- OTA_AirPriceLLSRQ (1.3.1)
- OTA_AirFlifoLLSRQ (1.1.1)
- OTA_AirBookLLSRQ (1.1.1)

Developer Resources

Support Procedures
SWS Web Service List
SWS Developer Start-up Kit
SWS Developer's Guide
WSE Topic Overview

Other Useful Sites

Sabre Holdings
Sabre eServices
Format Finder
The OpenTravel™ Alliance

https://webservices.sabre.com

E-Commerce

amazon webservices™

Browse Amazon Web Services

Amazon Web Services offers a variety of web services

Learn About Amazon Web Services

- AWS Home
- Why Use AWS?
- What's New in AWS?
- Upcoming Events
- Reference Applications
- Create an Account
- FAQs

Browse Web Services

- Amazon E-Commerce Service
- Amazon Historical Pricing
- Amazon Mechanical Turk (Beta)
- Amazon S3
- Amazon Simple Queue Service (Beta)
- Alexa Top Sites
- Alexa Web Information Service
- Alexa Web Search Platform (Beta)

Browse All Web Services

Amazon E-Commerce Service
Amazon E-Commerce Service (ECS) exposes
web site owners and merchants to leverage the data a
business. ECS 4.0 makes it extremely easy for develop

Amazon Historical Pricing
The Amazon Historical Pricing web service gives developers programm
books, music, videos, and DVDs (as sold by third-party sellers on Am
make informed decisions on pricing and purchasing.

Amazon Mechanical Turk (Beta)
Today, humans still significantly outperform the most powerful comp
objects in photographs – so
between human beings and
completed, and the compute
computer program could ask
human beings to perform a t

Amazon Mechanical Turk do
directly into their processin

Amazon S3™
Amazon S3 is storage for
provides a simple web servic
anywhere on the web. It giv
infrastructure that Amazon uses to run its own global network of wet

Sample Services:

- Retrieve product data
- Order a product
- Register your own product

Amazon also provides "Value add" services

S3 = "Simple Storage Service"

Creating a "Service Platform": Content + Utility + Context

Building a community be reducing effort to participate

http://aws.amazon.com/connect

Logistics

FedEx

Sample Services:

- Track a package
- Proof of delivery notification alert
- Courtesy rate quotes

Current emphasis on Internet & Http as transport

Provides client side libraries:

- Java, C++, Visual Basic

FedEx Ship Manager API Documentation

Documentation

- Overview (PDF) Describes the features and tools contained in FedEx Ship Mana… information about registering, integration, and certification. Also contains ref… help and documentation.
- FedEx API Client Libraries Developer's Guide (PDF) Describes the parameters of the function used to process transactions through the FedEx servers. The usage of both the standard C and COM interface is covered.
- Tagged Transaction Parser Developer's Guide (PDF) Describes the functions in the FSM API Parser Library and explains how to use them.
- Programmer's Label Guide (PDF) Contains instructions for using the Label Extensions Package for the preparation of Shipping Labels and Signature Proof of Delivery Letters.
- Error and Result Code Listing (PDF) Contains a comprehens… codes and their meaning. This covers all functions within the … Ship Manager API.
- Tagged Transaction Guide (PDF) Contains instructions for the… by FedEx Ship Manager API.
- XML Transactions Guide (PDF) Contains instructions for the X… Ship Manager API and FedEx Ship Manager Direct

FedEx Ship Manager® API/DIRECT

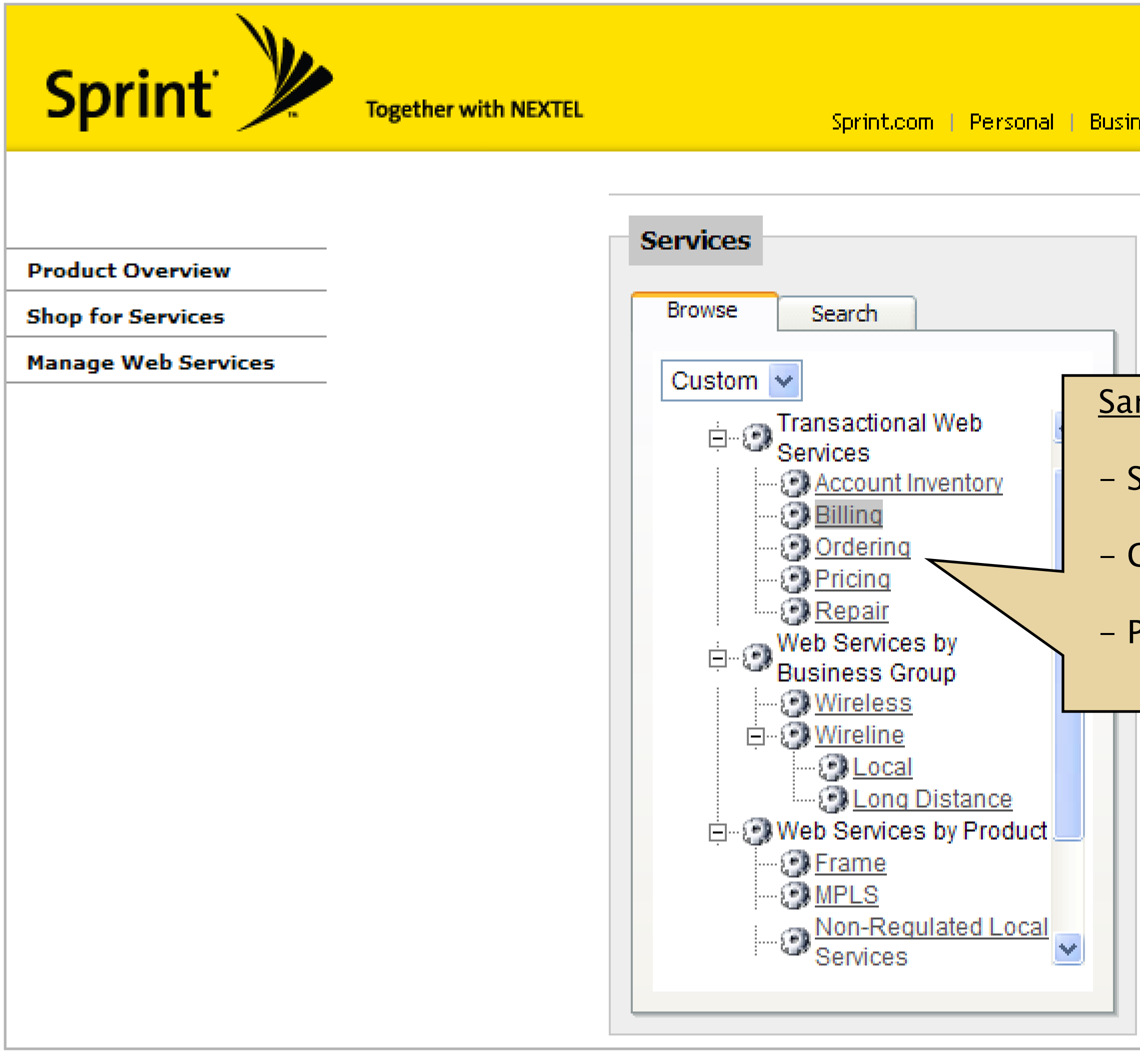
XML Transaction Guide

March 2006

270 pages of documentation!

http://fedex.com/us/solutions/shipapi/docs.html

Communications

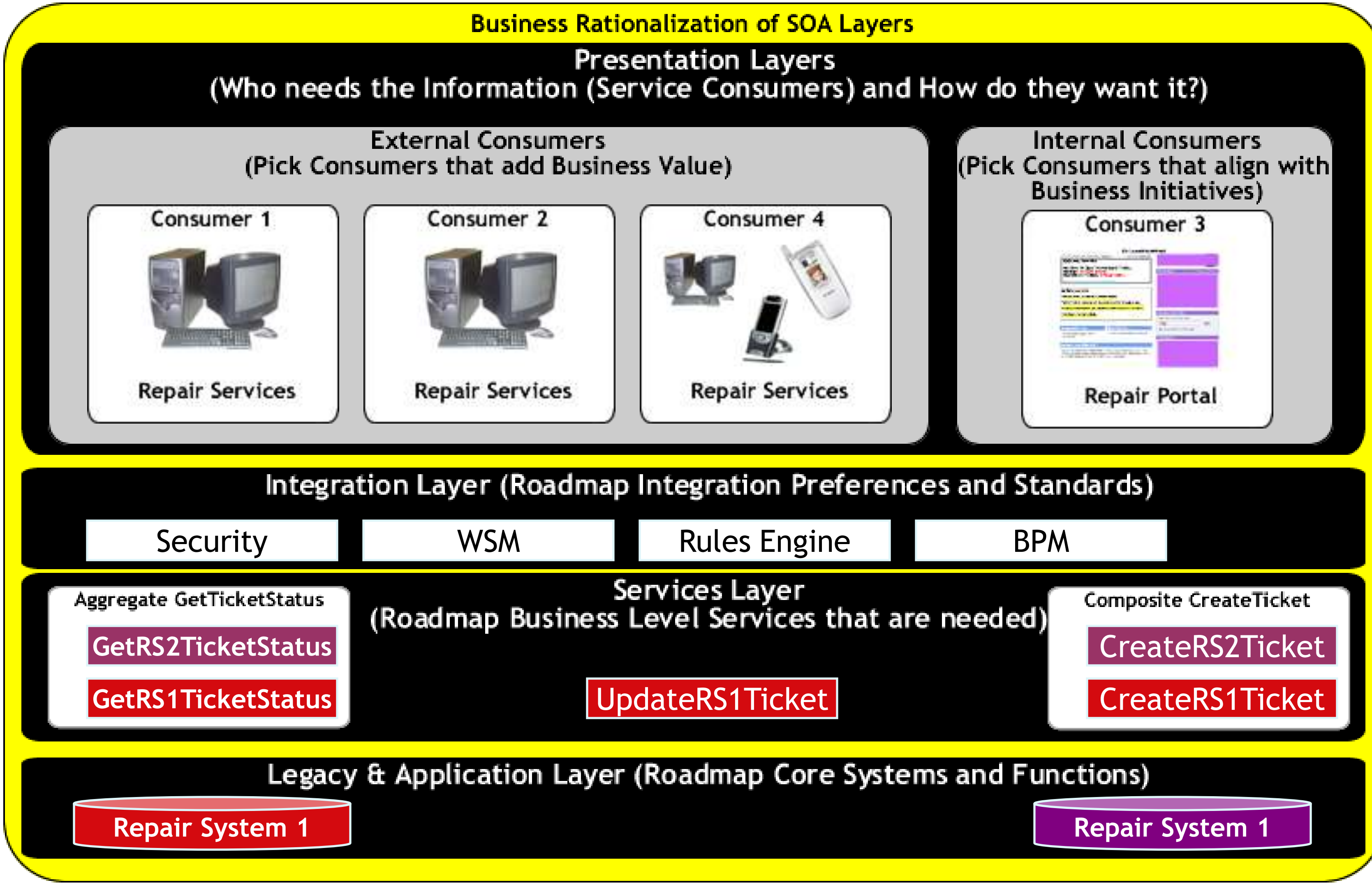


<u>Sample Services:</u>

- Schedule Repairs
- Check Inventory
- Price / Order

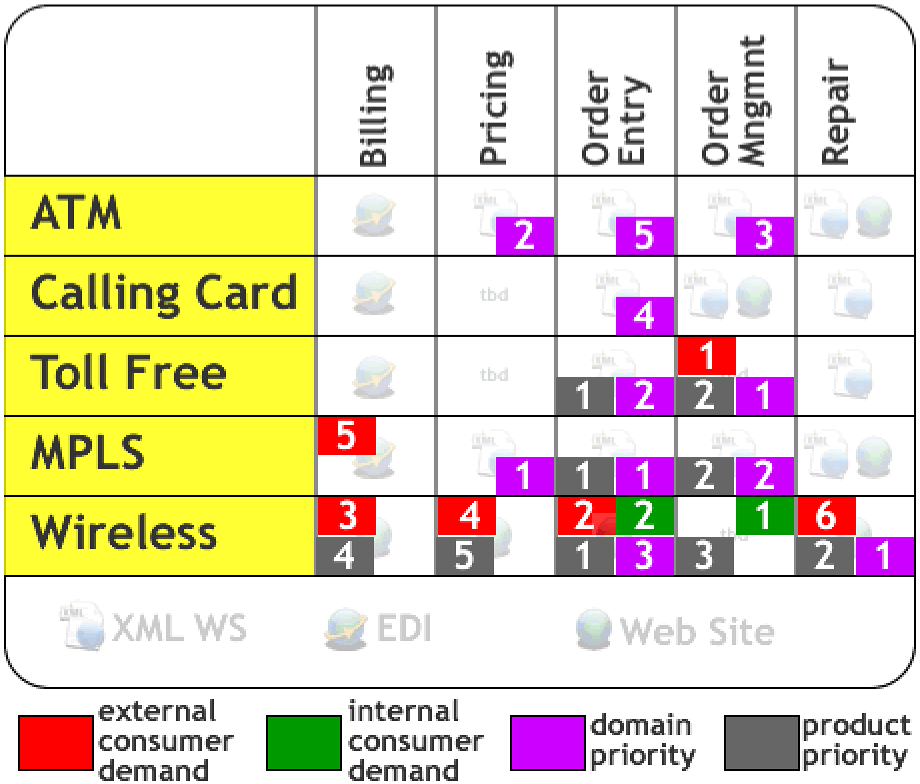
http://webservices.sprint.com/

The Sprint Reference Model



The Demand Model

Priorities and Demand Matrix



Create a demand matrix to determine what internal and external parties want.

The New Opportunity

Extending Services to Partners

Make it Happen!

- 7 SOA Opportunities:

1. Manageable
2. Synergy & Investment
3. User Centric Systems
4. Process Driven
5. Interoperability & Integration
6. Contracts & Service Levels
7. The Extended Enterprise

Our Offering

MomentumSI provides customers wanting to extend their systems reach to external parties a simple and cost effective solution.

- Define the ESN Center
- Leverage off-the-shelf technology
- Use proven solutions and patterns
- Tailor the environment to the unique needs of the client
- Ramp up organizational polices & practices
- Establish a holistic capability
- Get up & running quickly

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